Software Design and Analysis (CS3004)

Date: December 20th 2024

Course Instructor(s)

Engr. Abdul Rahman, Dr. Imran Ali Bhatti, Ms Sobia Iftikhar, Ms Javeria Farooq, Ms Syeda Rubab Manzar, Ms Iqra Fahad, Ms Fizza Aqeel

Final Exam

Total Time (Hrs):	15 mins
Total Marks:	20
Total Questions:	1

22K-4484 Roll No   SF Section   Student Signature

Don't write below this line. Return the paper after 15 minutes to the invigilator. Write answer only in tables given below the questions on this sheet. Cutting or overhand writing will result in negative marking.

CLO # 1: Understand the role of design and its major activities within the software development process, with focus on the Unified process.

Question 1:

(Part A) Choose only one correct option among the following on the table given below. **[1*10=10]**

1) Which UML diagram is used to represent the structure of a system?
(A) Sequence Diagram (B) Use Case Diagram (C) Class Diagram (D) Activity Diagram
2) What is a component in UML?
(A) A part of a class (B) A modular part of a system (C) A user of the system (D) An actor
3) Which diagram represents the flow of activities in a process?
(A) Class Diagram (B) Activity Diagram (C) Use Case Diagram (D) Component Diagram
4) What does an actor symbolize in a Use Case Diagram?
(A) External entity interacting with system (B) Class (C) Database (D) Software component
5) What is a package in UML?
(A) A collection of attributes (B) A container for grouping related elements (C) A type of actor (D) A part of a sequence
6) Which diagram is used to represent interactions between objects?
(A) Use Case Diagram (B) Sequence Diagram (C) Class Diagram (D) Activity Diagram
7) Which diagram helps to identify the boundaries of a system?
(A) Sequence Diagram (B) Use Case Diagram (C) Class Diagram (D) State Diagram
8) What is the purpose of a Deployment Diagram?
(A) To show how hardware and software components are deployed (B) To describe object interactions (C) To show the lifecycle of a system (D) To define the flow of activities
9) What symbol represents a decision point in an Activity Diagram?
(A) Circle (B) Rectangle (C) Diamond (D) Arrow
10) Which UML diagram is used to represent modular parts of a system?
(A) Sequence Diagram (B) State Diagram (C) Component Diagram (D) Activity Diagram

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	B	A	B	B	C	A	C	C

(Part B) Write T (for true) or F for (False) on the table given below. [1*10=10]

1. UML stands for Unified Modeling Language.
2. A class in a Class Diagram is represented as an ellipse.
3. Sequence Diagrams show the interaction between objects over time.
4. A dependency in UML is represented by a dashed line with an open arrowhead.
5. An actor in a Use Case Diagram can be a user or another system.
6. The aggregation relationship in UML is shown with a hollow diamond at the source end.
7. UML diagrams cannot be used to represent software architecture.
8. A lifeline in a Sequence Diagram represents an object or participant in the interaction.
9. A state machine diagram represents the lifecycle of a single object.
10. Activity Diagrams are used to model the workflow of a system.

Question	1	2	3	4	5	6	7	8	9	10
Answer	T	F	T	T	T	T	F	T	T	T

18/20

Software Design and Analysis (CS3004)

Date: December 20th 2024

Course Instructor(s)

Engr. Abdul Rahman, Ms Fizza Aqeel, Dr. Imran Ali Bhatti, Ms Sobia Iftikhar, Ms Javeria Farooq, Ms Syeda Rubab Manzar, Ms Iqra Fahad.

Final Exam

Total Time (Hrs): 165 mins

Total Marks: 70

Total Questions: 2

22k-4596 Roll No

F Section

Waleed Student Signature

Do not write below this line.

CLO # 2: Design and Implement OOD models and refine them to reflect implementation details.

Question 1: UML modeling and Analysis [50 points]

Question 1 (A): ECB class diagram [10 points]

A multinational e-commerce company, GlobalShop that operates an online marketplace where users can buy and sell products in various categories, including electronics, clothing, home goods, and more. The platform is available in multiple countries, with different regional warehouses, currencies, and payment gateways. The company provides a marketplace for both customers and sellers while also having internal staff such as administrators. The platform ensures compliance with local laws, supports multiple languages, and integrates with international shipping and payment systems. GlobalShop offers the following features:

1. Customer Portal: Customers browse products, make purchases, manage shopping carts, track orders.
2. Seller Portal: Sellers can list products, update inventory, manage orders, and view customer feedback.
3. Admin Portal: Administrators oversee the entire platform, manage users, handle customer service, and ensure the platform adheres to company policies and local regulations.
4. Payment System: Manages transactions, supports multiple currencies, and integrates with various payment gateways to facilitate purchases globally.

The key entity resources that contain and process information are: Customer that represents individuals who purchase products on the platform. A customer has Customer ID, order history, payment methods, and shipping addresses. Seller that represents businesses or individuals who sell products on the platform. A seller has some data attributes such as Seller ID, product listings, order history, inventory data. Product that represents products listed by sellers on the platform. Product has the attributes Product ID, product name, description, price, stock quantity. Payment which represents the payment for an order. A payment can have the attributes such as Payment ID, amount, currency, payment method, transaction status and the Admin that represents the internal administrators who manage the platform. Here Admin has attributes such as admin ID, user permissions, and access logs.

The controllers represent the business logic that allows users to interact with the system and provide the mechanisms that handle data operations for each role. Customer Controller is responsible for managing customer activities like placing orders, managing shopping carts, tracking shipments, and interacting with customer support. Seller Controller handles seller operations such as adding products, updating inventory, managing orders, and viewing customer feedback. Admin Controller manages the overall system, moderates content, manages user accounts, processes disputes, and ensures compliance with regional laws. Payment Controller manages transactions, handles currency conversion, and ensures secure payment processing.

In this scenario boundaries define the access limits between entities and controllers to ensure each user role only has access to their relevant resources. Such as Customer Boundary can access personal order history, shopping cart, payment methods, product details, and shipping information. A Seller can have the Access to product listings, order history for their own products, inventory management tools, and customer feedback. Admin can access to all platform resources, including user management, seller activities, customer transactions, payment logs, and platform settings. A Payment Boundary can Access the Transaction details, payment amounts, and currency information, but only for their own orders or sellers' transactions.Draw ECB diagram (use stereo types in a regular class diagram. Show methods and attributes as well)

NOTE: In your analysis, you need to take care of all kinds of cardinalities in UML class model, be careful to think about all possible kinds of relationships between different use cases and classes. Think of all possible elements of UML modeling that you can infer from the given scenario, however do not overthink to extend the requirements of the given scenario, just analyze what is provided to you in the scenario.

Question 1(B):UML Real time Activity model ✓ **[10 points]**

A Ticket Vending Machine (TVM) automates the process of purchasing tickets for public transportation. The process involves several steps, interactions, and decision points:

User Interaction:The user starts by selecting their preferred language on the TVM screen.The user selects the type of ticket: single ride, return ticket, or multi-day pass. If the user chooses a return ticket, the system asks for the return time and date.The user enters their destination or selects it from a list of predefined stations. Pricing and Options: The system calculates the ticket price based on the user's selection and displays the amount.

If the user qualifies for a discount (e.g., senior citizen, student), they can apply it by scanning a valid ID or entering a promo code.Payment Process:The user selects a payment method: Cash: The machine accepts the required amount and returns change if applicable. Card: The user inserts their card, and the machine processes the payment via the bank. Digital Wallet: The user scans a QR code to complete the payment using a mobile wallet. If the payment fails, the system prompts the user to retry or cancel the transaction. Ticket Printing Once payment is successful, the system prints the ticket(s). If the user selected a multi-day pass, the ticket includes specific details like validity period and usage instructions. Concurrently, the system updates the backend database to log the transaction and ensure the ticket's barcode is registered for validation. If the printer malfunctions, the system notifies the user and refunds the payment. If the machine runs out of paper or coins for change, the system stops new transactions and alerts the maintenance team. After receiving the ticket, the user can optionally print a receipt. The system resets and is ready for the next transaction. Create an Activity Diagram for a "Ticket Vending Machine" (TVM) Process based on the detailed scenario below. Your diagram must include Swimlanes, Forks, Joins, and Decision Points to demonstrate the process comprehensively. Ensure the workflow reflects interactions between the user, machine, and backend systems. Ensure a logical workflow that begins with language selection and ends with ticket printing and user satisfaction.

Question 1(C):UML State Chart model **[10 points]**

The following scenario describes the lifecycle of a subscription account in an online movie streaming service. The lifecycle starts when a user registers on the streaming platform by creating an account. The registration process involves entering personal details, choosing a subscription plan, and setting up payment methods. Once registration is complete, the subscription becomes *Active*. In this state, the user can stream movies, manage their watchlist, and update their profile or payment details. The subscription remains active as long as payments are made on time. If the subscription payment is not received by the due date, the account transitions to the *Payment Pending* state. Notifications are sent to the user reminding them to make the payment. If the payment is not received within a grace period, the account becomes *Suspended*. The user loses access to streaming services but can still log in to view account details and resolve payment issues. If the payment is made while the account is in the *Payment Pending* or *Suspended* mode, the account becomes active again. If the user cancels the subscription or does not resolve payment issues after an extended period, the account transitions to the *Canceled* state. Once canceled, the account remains in this state and can only be reactivated by creating a new subscription. The account may transition to the *Terminated* state if the user permanently deletes it or if the system terminates it due to violations of terms of service. In this state, all account data is deleted, and the user cannot reactivate the account. Design a UML State Chart Diagram that represents the lifecycle of the subscription account object in the online movie streaming service. Ensure your diagram clearly shows all possible states and transitions between them based on the scenario.

Question 1(D):UML Component model **[10 points]**

The **Smart Home Energy Management System (SHEMS)** is a comprehensive solution designed to optimize energy usage in residential buildings by integrating various subsystems. These subsystems include **Energy Consumption Monitoring**, **Smart Appliances**, **Energy Storage Management**, **Renewable Energy Integration**, **User Interface**, and **Demand Response Management**. The system aims to reduce energy wastage, lower costs, and enhance sustainability.

The **Energy Consumption Monitoring System (ECMS)** tracks real-time energy usage via energy meters and smart plugs, sending data to the **SHEMS Central Controller** for processing. This data helps optimize energy consumption, and the insights are displayed to the user via the **User Interface**. The **Smart Appliances System (SAS)** includes smart thermostats, lights, and appliances, which adjust their behavior based on energy optimization goals, sending usage data to the monitoring system. The **Energy Storage Management System (ESMS)** manages home battery storage, ensuring efficient charge and discharge cycles to store excess energy for later use. It works in conjunction with the **Renewable Energy Integration System (REIS)**, which incorporates solar panels and wind turbines to generate renewable energy. This renewable energy is sent to the **SHEMS Central Controller** and can be stored or used based on demand.

The **User Interface System (UIS)** allows homeowners to control energy usage, monitor reports, and manually adjust appliance settings through a mobile app or dashboard. The **Demand Response Management System (DRMS)** communicates with the utility grid, adjusting energy usage during peak demand periods by turning off non-essential appliances, and providing incentives for energy conservation.

The **SHEMS Central Controller** serves as the brain of the system, integrating data from all subsystems to optimize energy usage, storage, and renewable energy integration. Through its interfaces with the various subsystems, the controller ensures seamless coordination and efficient operation of the entire system. Overall, SHEMS provides a sustainable, cost-effective way to manage energy consumption in modern homes.

Question 1(E):UML Deployment model **[10 points]**

In a sophisticated deployment diagram for a Netflix-like streaming platform, we depict a dynamic, multi-tiered architecture designed to handle millions of simultaneous users with exceptional speed and reliability. At the forefront, **user devices**—smartphones, smart TVs, tablets, and laptops—interact seamlessly with **load balancers** that distribute incoming traffic across multiple **web servers** to ensure scalability and fault tolerance. These **web servers** act as the gateway to the system, processing user requests, managing sessions, and providing initial content delivery through the **content delivery network (CDN)**. The CDN optimizes streaming by caching videos across strategically located **edge servers**, reducing latency and improving the quality of experience for users globally.

Behind the scenes, **application servers** execute the core business logic, from user authentication and recommendation algorithms to subscription management and content personalization. These application servers communicate with a cluster of **database servers** to handle vast amounts of user data, preferences, and real-time analytics. For high availability, the database architecture employs **replication and sharding**, ensuring quick access to data while balancing the load.
Video storage servers, distributed across multiple data centers, store massive media libraries. High-performance **object storage** systems, such as Amazon S3 or custom solutions, are integrated to manage the vast scale of video content. To support seamless streaming, **transcoding servers** continuously convert videos into multiple formats and resolutions, catering to different bandwidths and device capabilities.
Additionally, **monitoring and logging servers** track system health, performance, and user activities, sending real-time alerts to ensure any issues are quickly mitigated. The deployment diagram would also account for **microservices**, running independently in containerized environments (e.g., Kubernetes clusters), each responsible for discrete functionalities like billing, content management, and user recommendations, all working in harmony to deliver a high-performance, highly resilient platform that adapts to millions of requests and an ever-expanding content library.

CLO # 3: Apply and use UML to visualize and document the design of software systems.

Question 2: Design Patterns **[20 points]**

You are tasked with integrating a new online flight reservation system with an existing legacy booking system. The new system supports multiple airlines and allows users to book flights using a variety of payment methods. However, the legacy system only supports a specific airline and payment method, and it does not provide direct access to its data. The challenge is to make the new system compatible with the legacy system so users can book flights seamlessly.

a. Identify the pattern which best suits the above scenario.
b. Write down the java source code for your selected design pattern based application.
c. Write some useful and sensible driver code in the main entry point of application that will demonstrate proper and complete functionality of the application.
d. Draw the diagram for your selected design pattern

Note: If you identify wrong pattern, then answers to all subsequent parts will also be invalid. Demonstrate. In this scenario, you need to also make some assumptions of the proper objects and their attributes and behaviors. Your assumption must not contradict with the scenario. Marks will be granted on the proper analysis of the system and its implementation.

- target => what client wants
- Adapter => to give what client wants
- Adaptee => the old which client dont want